EE 3EY4
Lab 4

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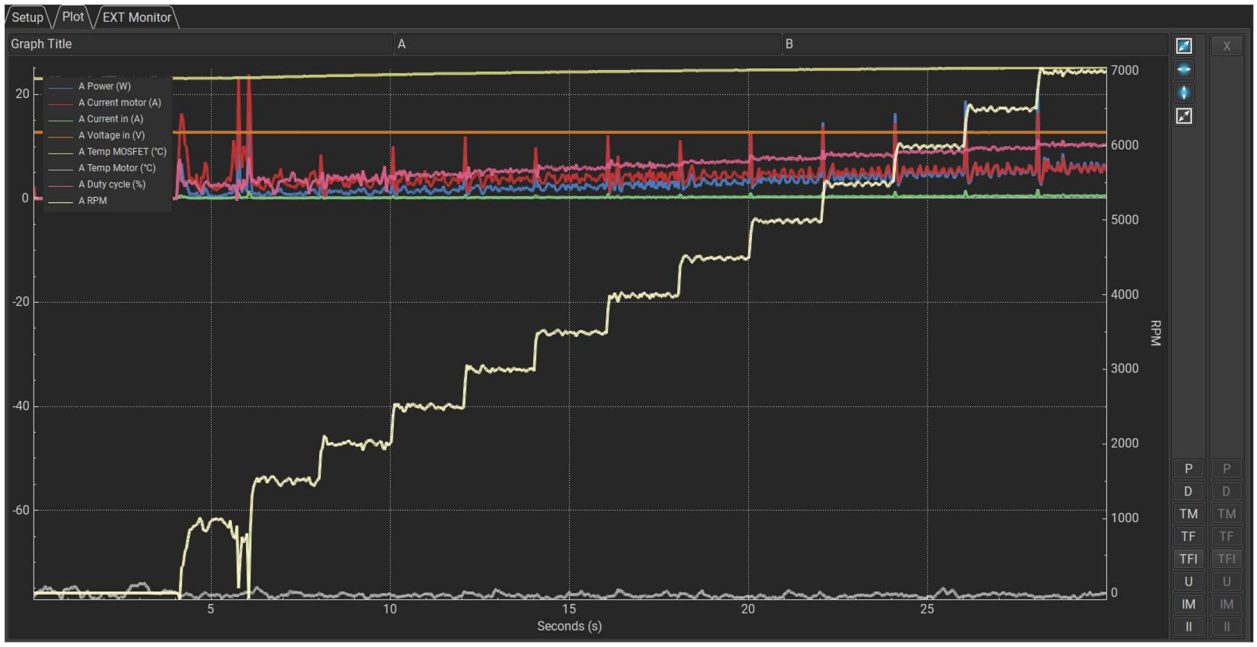
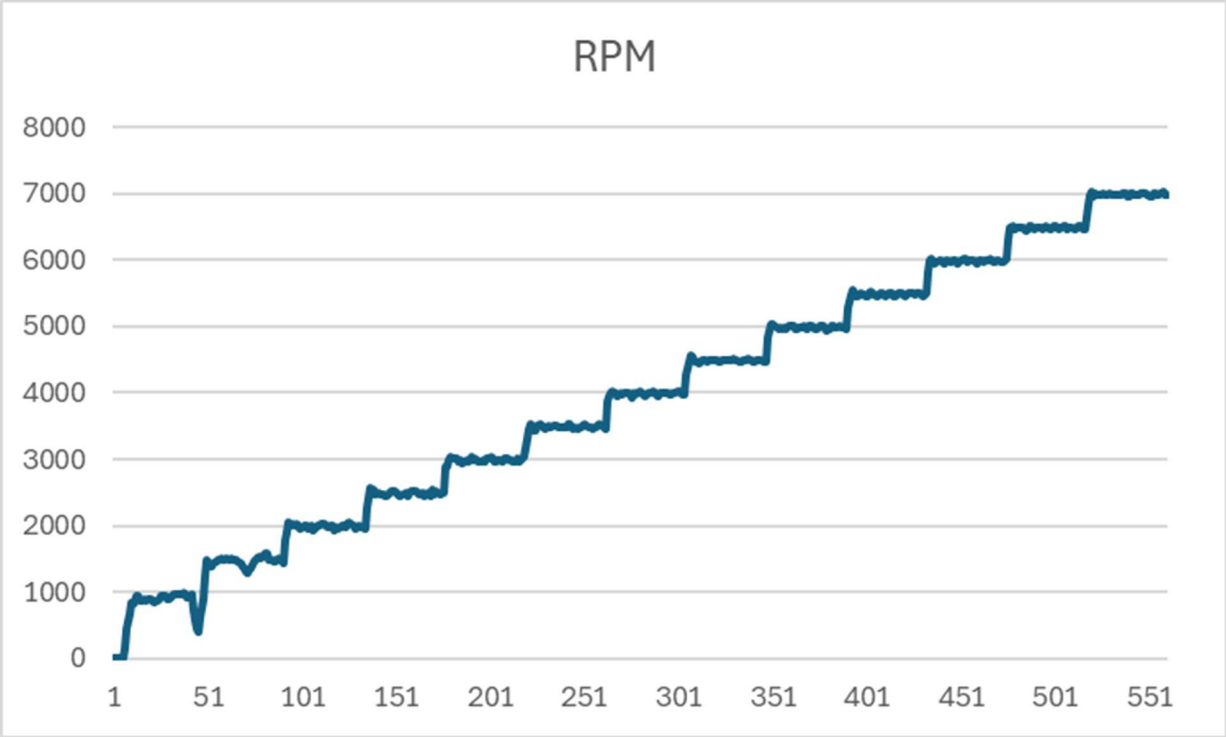
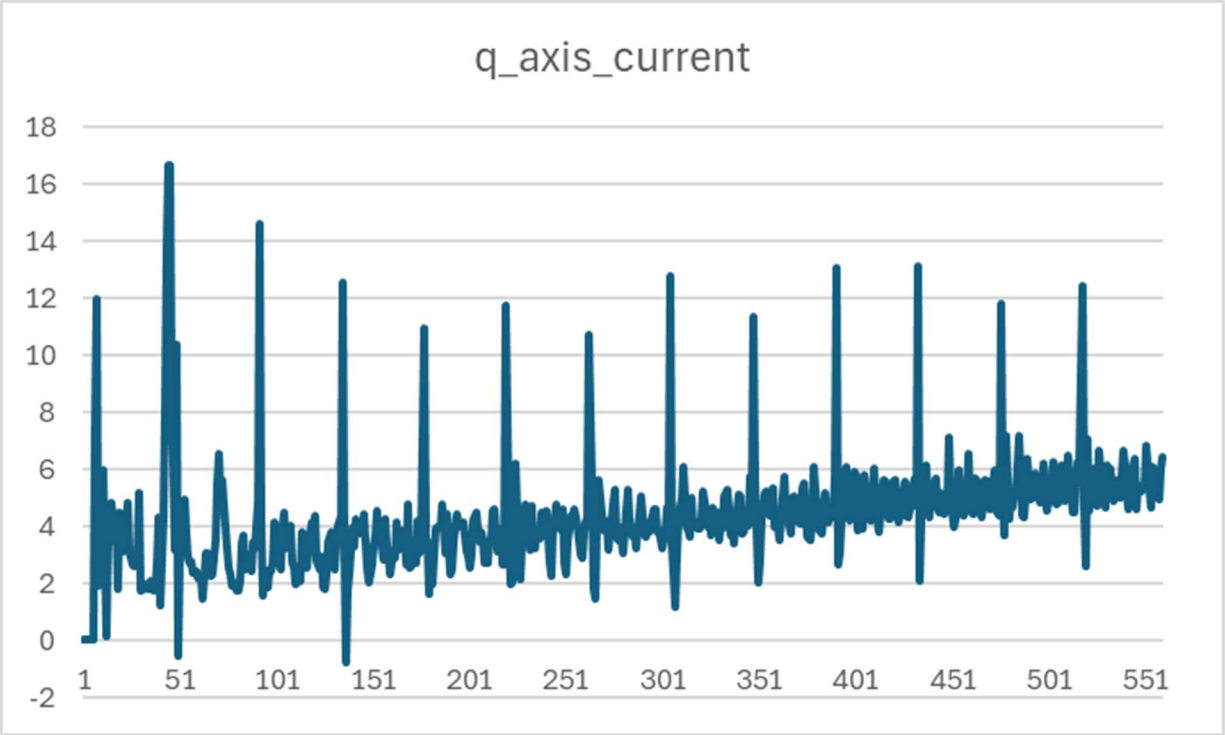
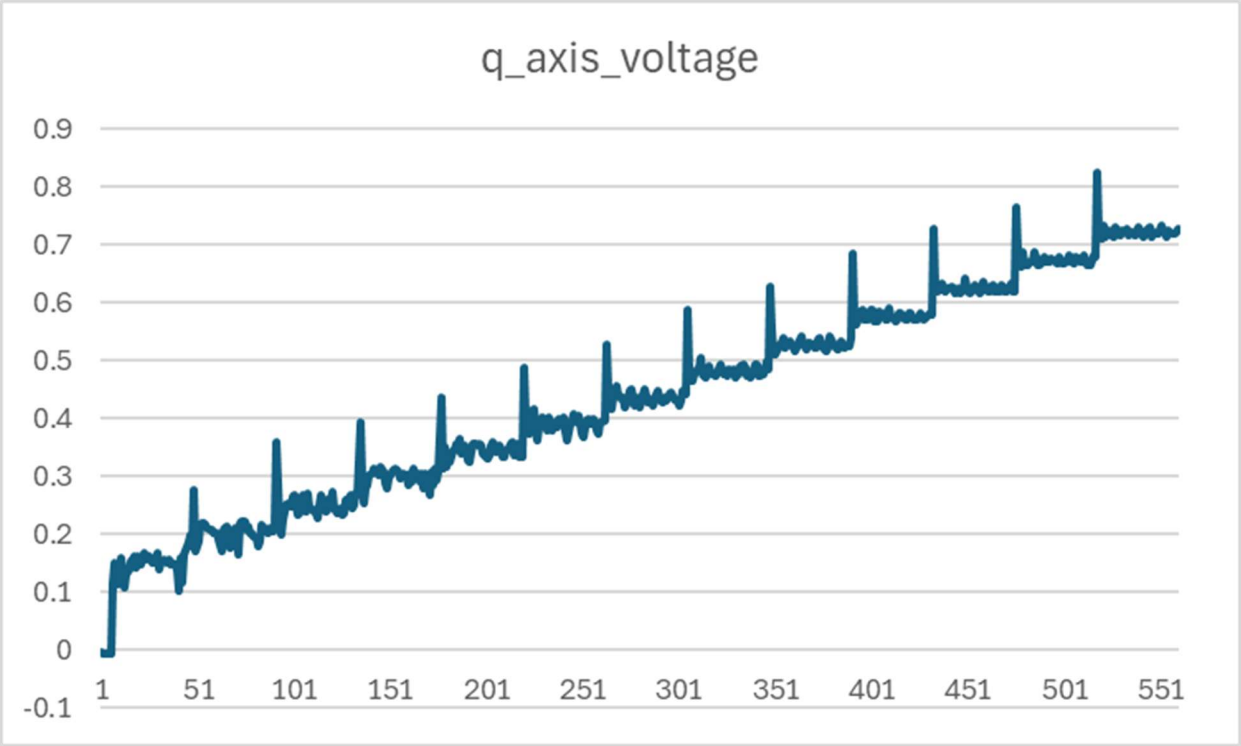
Objective 2:

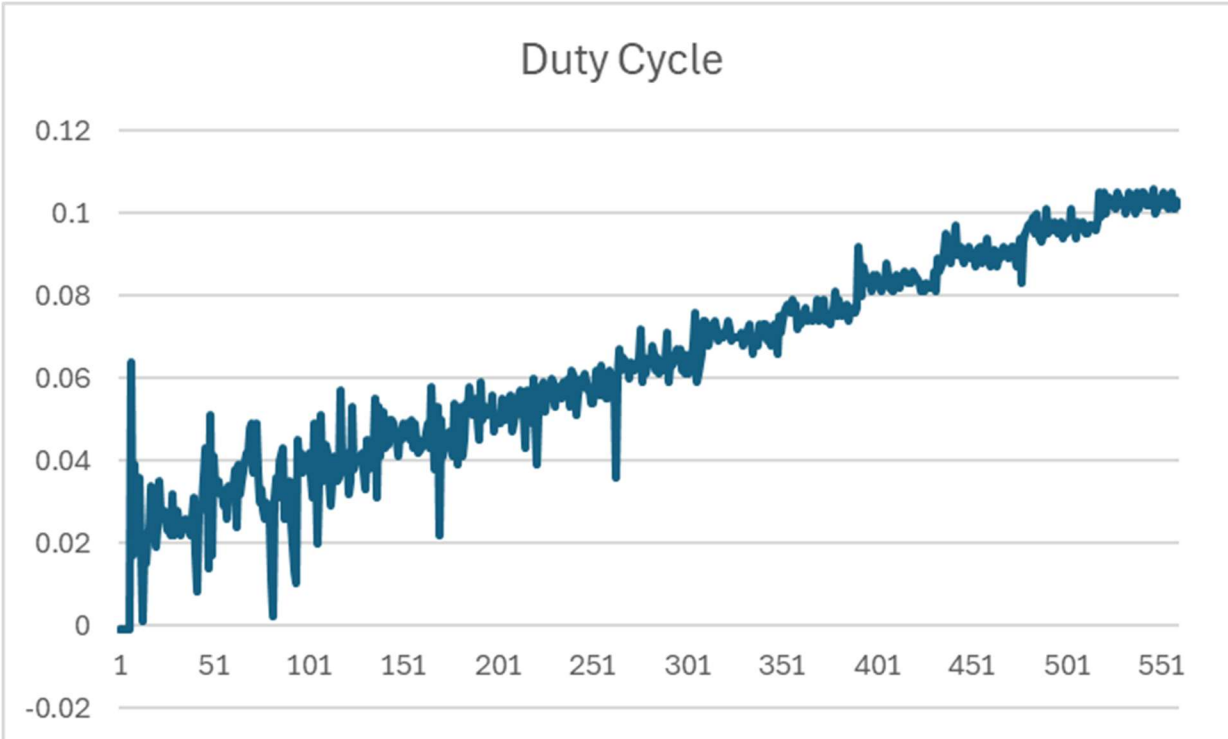
Question 1: Record the experiment results in a .csv file and plot them for your report. Explore how you can log the experiment with Realtime Data feature in the VESC Tool. Attach a screenshot of your plot tab in your report. Attach four graphs from your .csv file: one for RPM, one for q-axis current, one for voltage, and one for duty cycle. Explain how these variables change as the speed increases. What is the relationship between duty cycle, voltage, current, and speed?






q_axis_current
18
16
14
12
10
8
6
4
2
0
-2
1
51
101
151
201
251
301
351
401
451
501
551

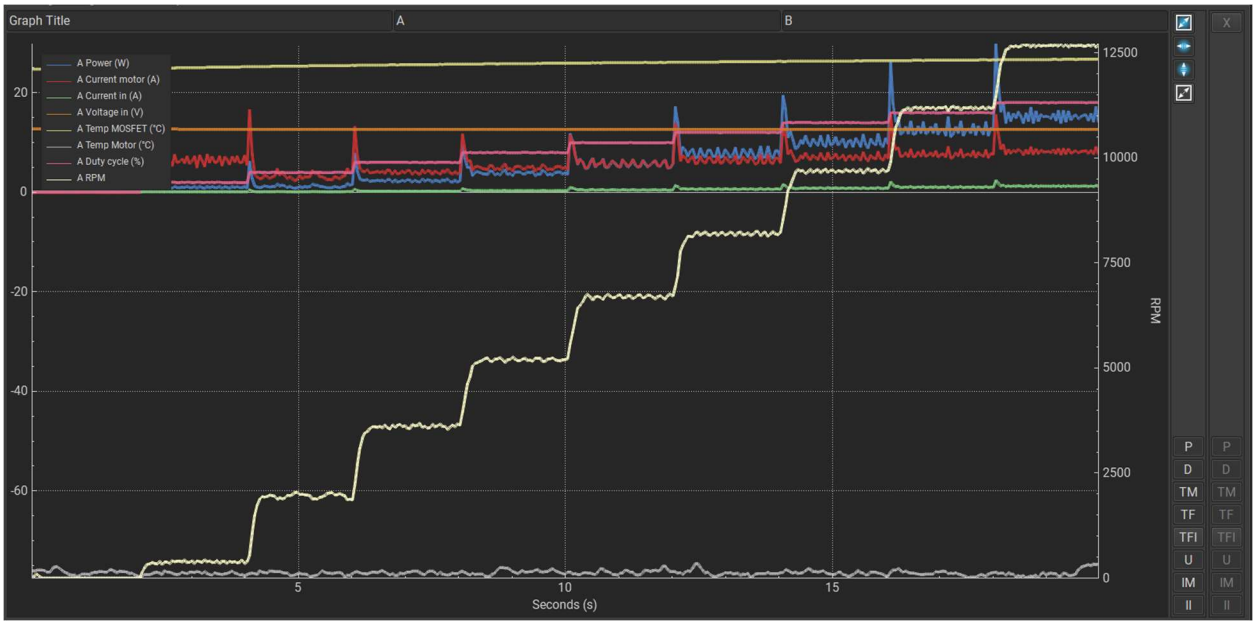

q_axis_voltage
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
-0.1
1
51
101
151
201
251
301
351
401
451
501
551

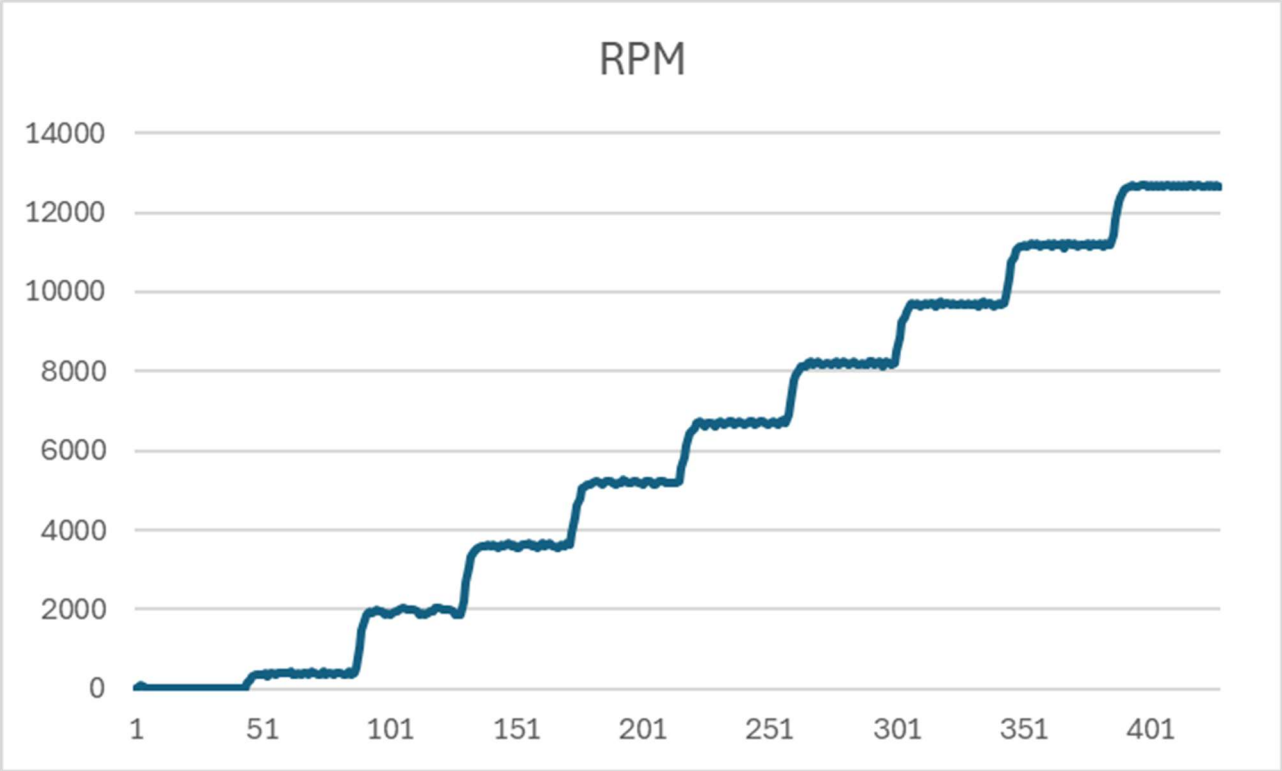


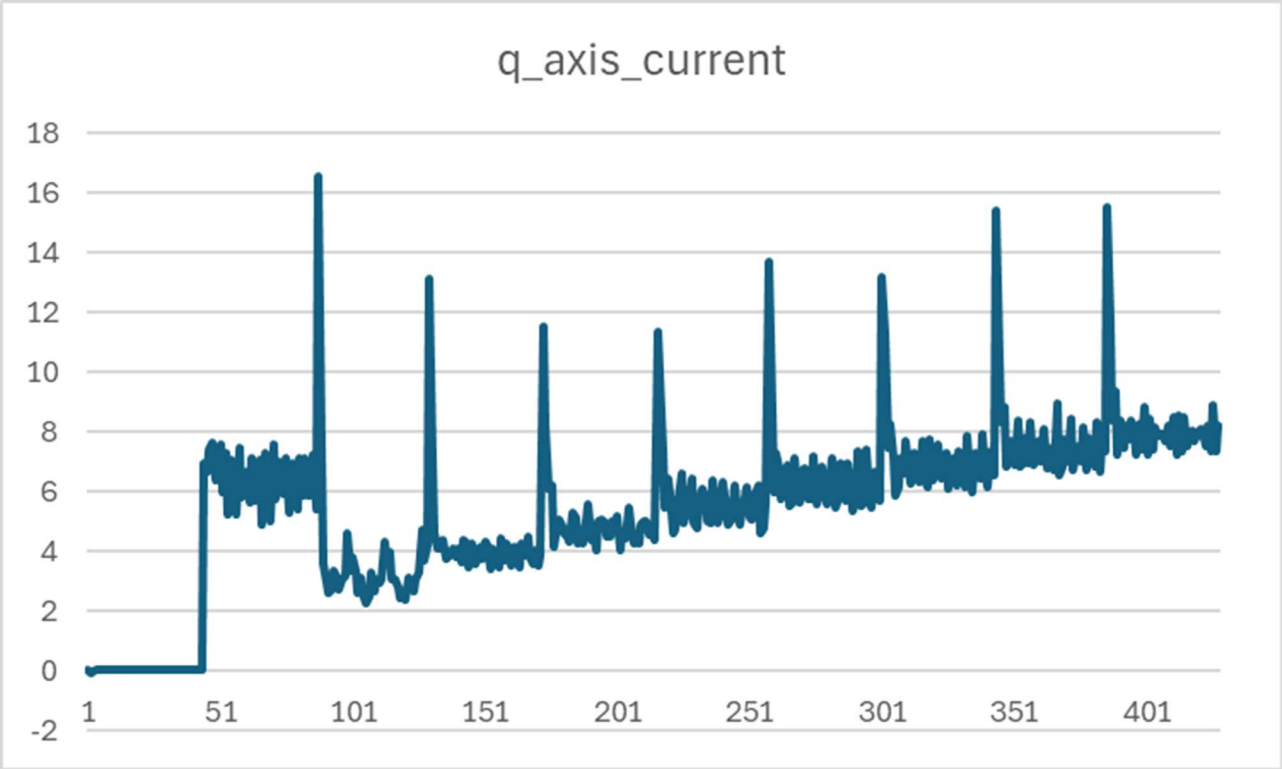
As the speed increases, the RPM increases in distinct steps, indicating that the sweep speed command uses steps to increment the speed. After each increase, the RPM settles for a little bit before incrementing again, indicating that the closed loop control is indeed working. By analysing the graph for q_axis_current we can deduce that the trend is that the average q-axis current slowly increases as the RPM increases. Periodically, there are also big spikes that occur at each increment of speed in the q_axis current which occurs as the current is inversely proportional to the torque, which results in a spike in current when the motor requires extra torque to accelerate when there is an increase in speed. Afterwards, the torque decreases, which lets the current settle down into a more constant value. The graph for the q_axis voltage looks like the graph for RPM as it also discretely increases with speed but with small spikes at every speed step. The graph for the duty cycle is more linear and roughly follows the increasing trend of the speed. There is a lot of noise at lower speeds for the graph of the duty cycle, but it subsides at higher speeds.

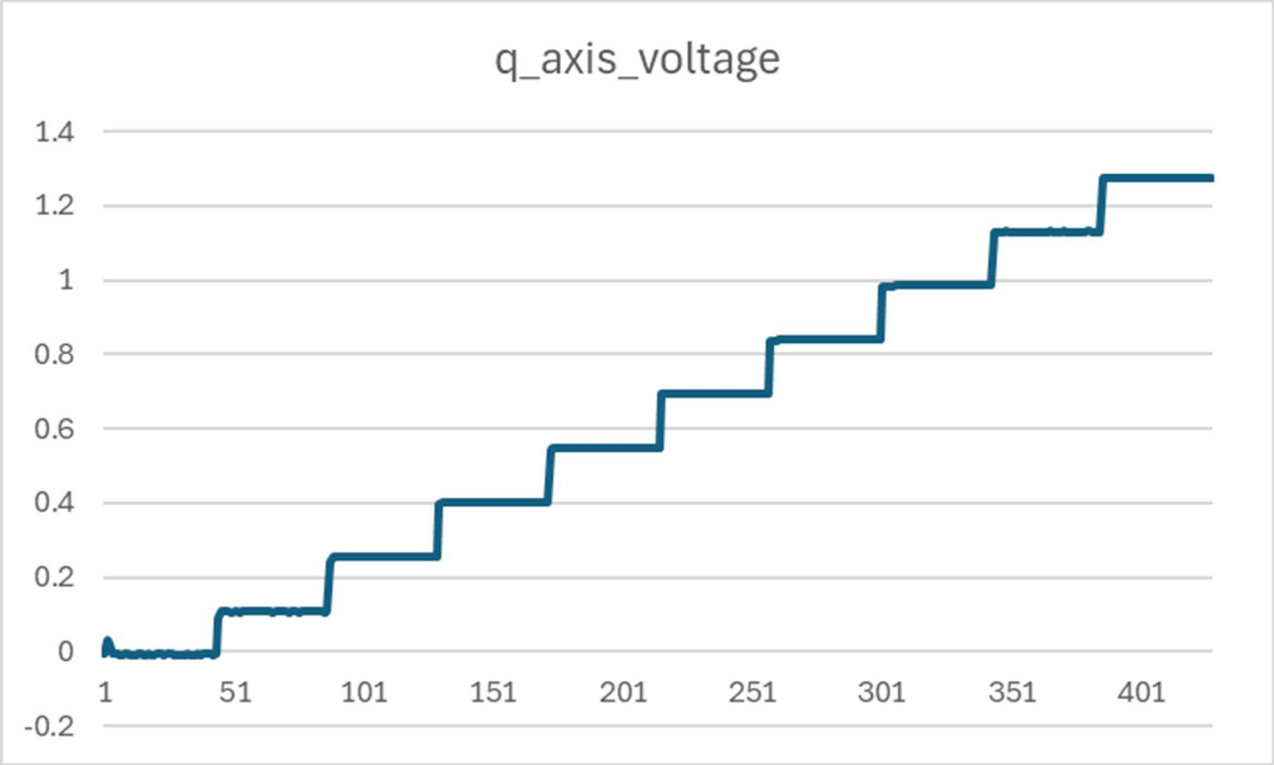
Overall, the relationship of the four parameters is that as the duty cycle is increased, the q-axis voltage is also increased to counteract the back EMF, while the q-axis current spikes during the acceleration due to the torque.

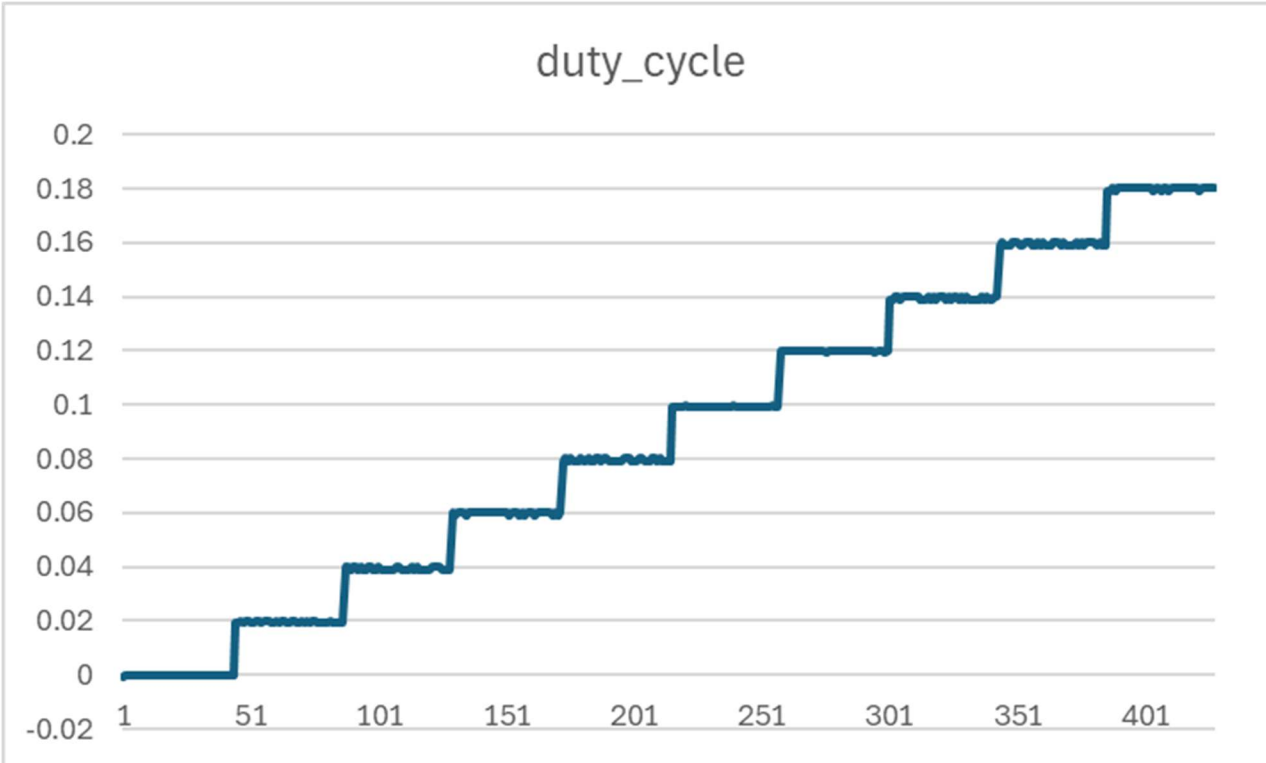
<u>Question 2:</u> Conduct a similar experiment and sweep the duty cycle command. The maximum duty cycle should be 0.2. Record the speed, q-axis current, voltage, and duty cycle. Plot the result. Compare the results with those from Question 1 when the speed command was swept. What is the difference in speed, current, and voltage response? Explain the reason why.


Graph Title
A
B
A Power (W)
A Current motor (A)
A Current in (A)
A Voltage in (V)
A Temp MOSFET (°C)
A Temp Motor (°C)
A Duty cycle (%)
A RPM
20
0
-20
-40
-60
12500
10000
7500
5000
2500
0
RPM
5
10
15
Seconds (s)
P
D
TM
TF
TFI
U
IM
II


RPM
14000
12000
10000
8000
6000
4000
2000
0
1
51
101
151
201
251
301
351
401


q_axis_current
18
16
14
12
10
8
6
4
2
0
-2
1
51
101
151
201
251
301
351
401


q_axis_voltage
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
1
51
101
151
201
251
301
351
401



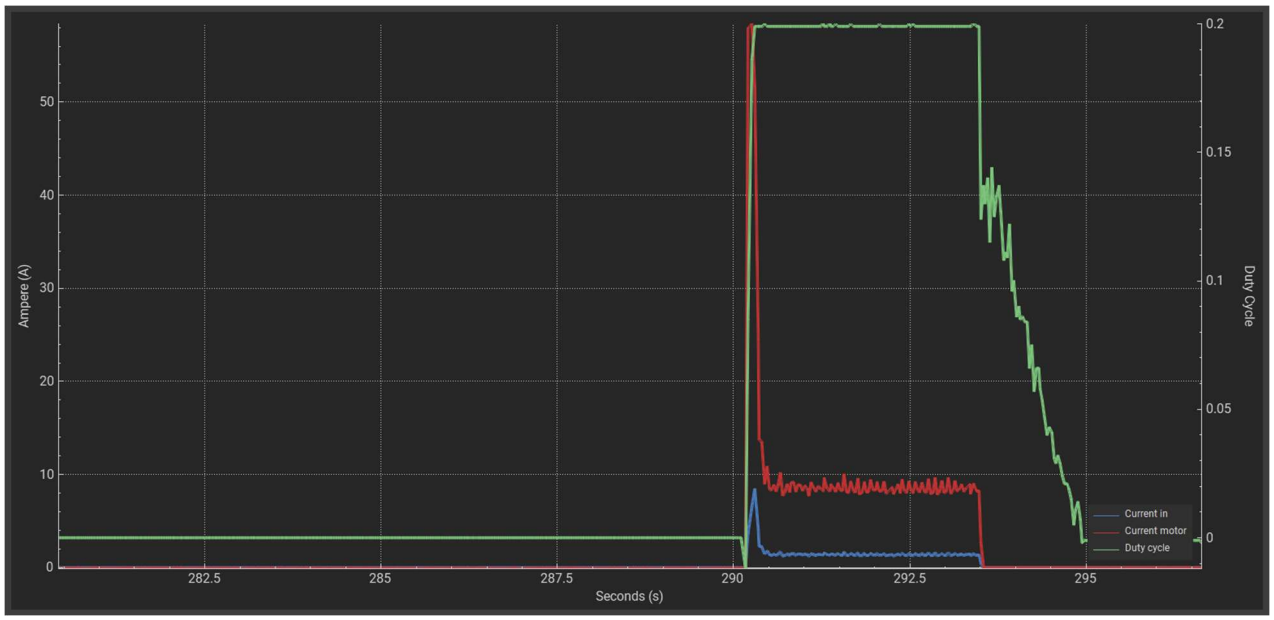
Compared to the results found in question 1, the graph for RPM follows the same trend as before but the increasing steps for the RPM are larger but less frequent, with the steady state being a smoother line than what was observed in question 1. The q_axis current in this question also follows a similar trend to the one found in question 1, but the spikes found at every step increase is smaller, with the general linearly increasing trend being slightly steeper. Similarly, the q_axis voltage follows the same general trend as the RPM and the q_axis voltage graph from question 1, where it increases discretely, but in this case, the steps are much clearer and cleaner. The duty cycle graph, however, differs the most as in this part, the graph looks a lot like the RPM graph which increases discretely, compared to the very noisy but roughly increasing trend of the graph found in question 1.

The reason for the differences in speed, current, and voltage responses are due to the increases in the speed command being larger but having less frequent steps, giving more time for the system to settle after each increase. This is the reason for the smoother graph for the RPM, as well as smaller spikes in the q-axis current as there is lower instantaneous acceleration, and a cleaner, more discrete voltage response as the increase in voltage is more controlled when counteracting the back EMF.

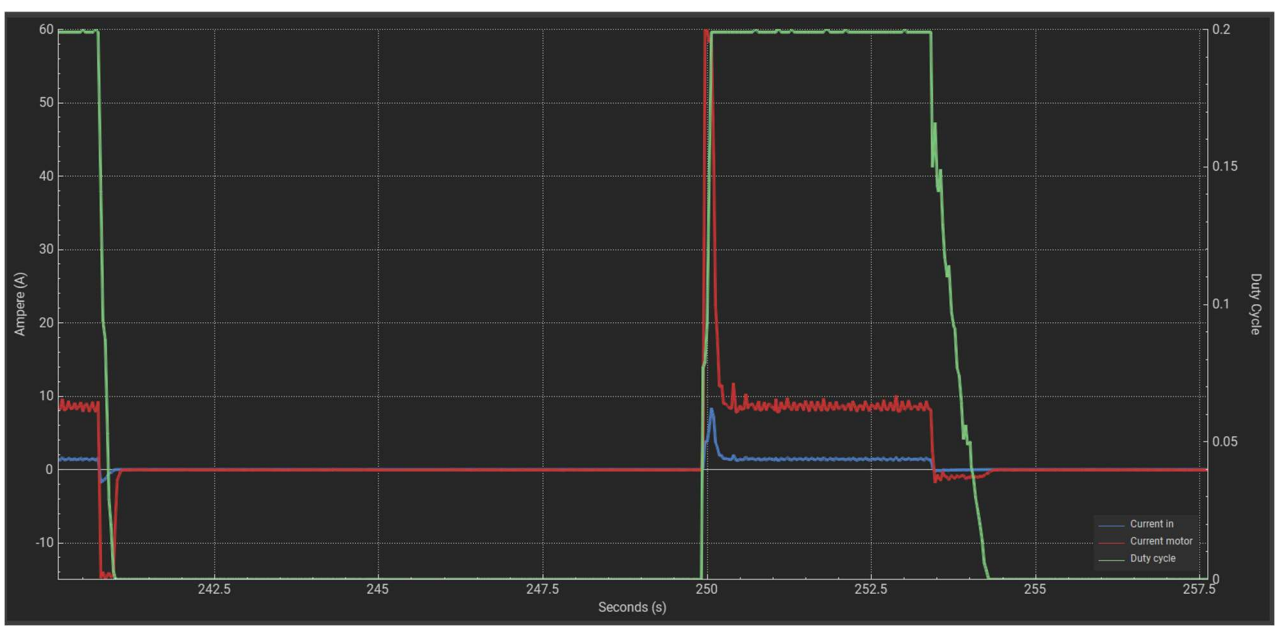
Question 3: Conduct an experiment where you operate the motor at 0.2 duty cycle. When the motor speed reaches steady state, stop the motor operation using STOP button and record the real-time speed, current, and voltage waveforms. Then conduct the same experiment and, when stopping the motor, apply brake current command (IB) of 1A. Conduct the same experiments multiple times and increase the break current gradually up to 15A. Record the data. How does

the speed response change? How does the q-axis current change? Explain the reason in the changes using the operating principles of the motor.

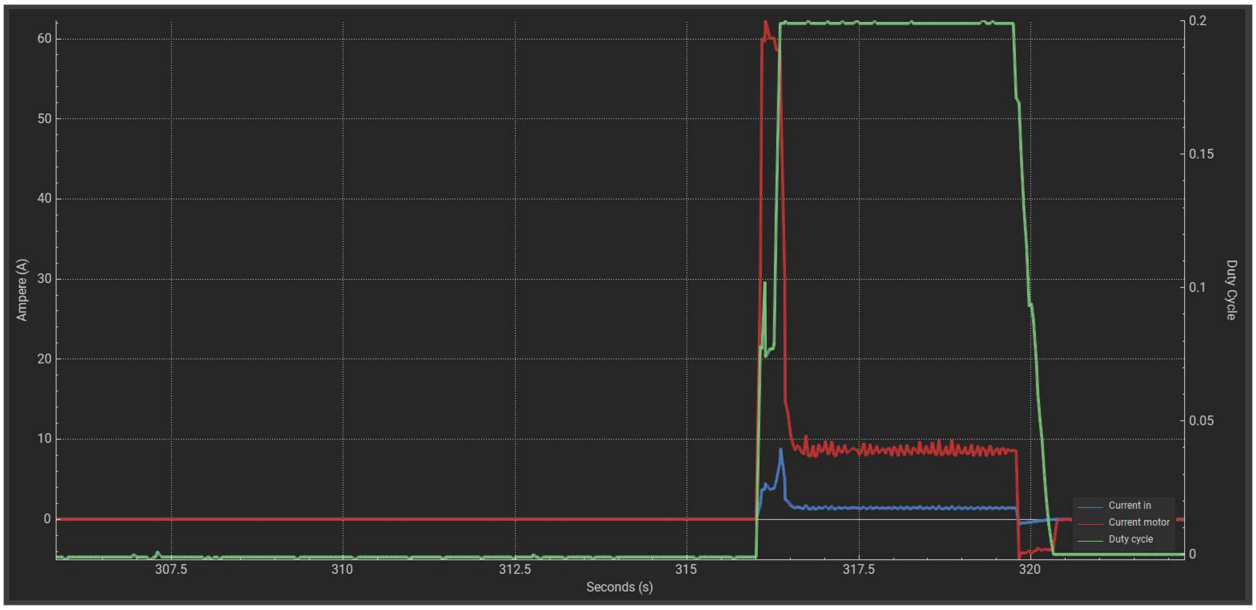
STOP



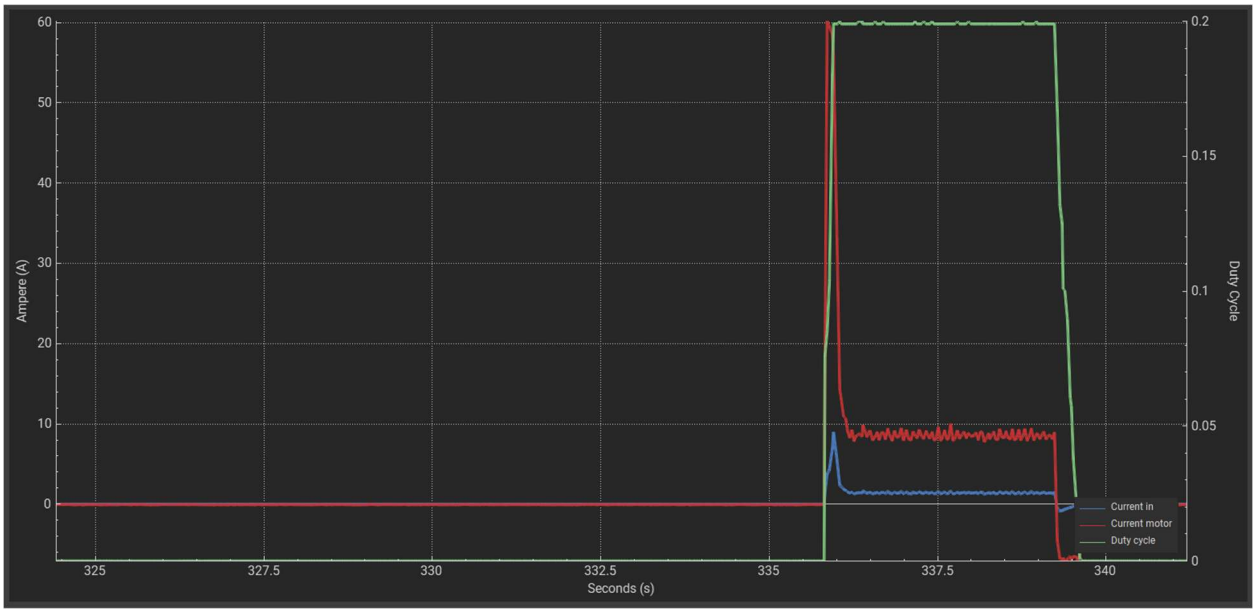
1A



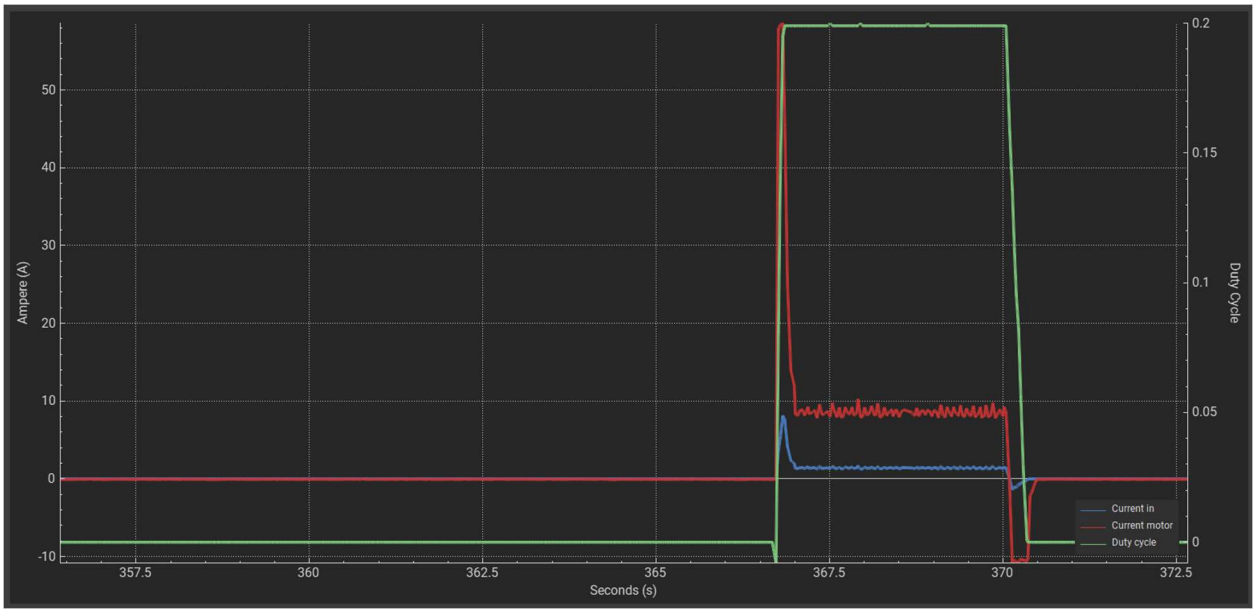
4A


Ampere (A)
Duty Cycle
Seconds (s)
Current in
Current motor
Duty cycle

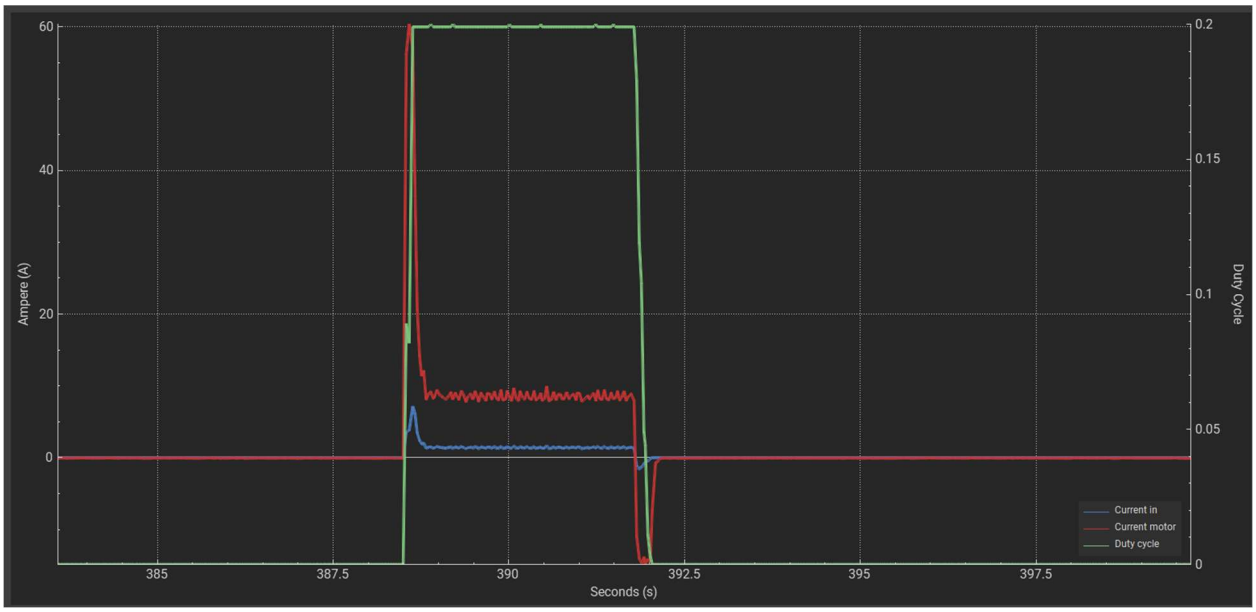
7A

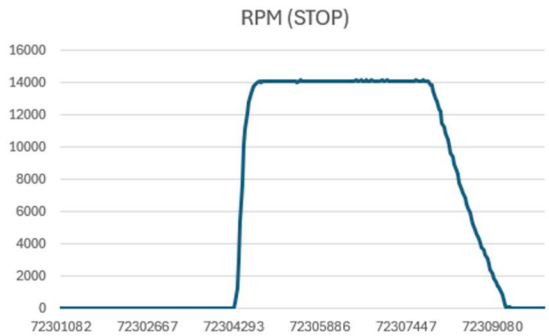

Ampere (A)
Duty Cycle
Seconds (s)
Current in
Current motor
Duty cycle

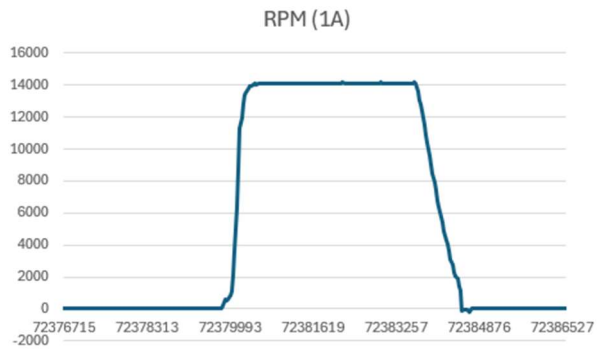
11A

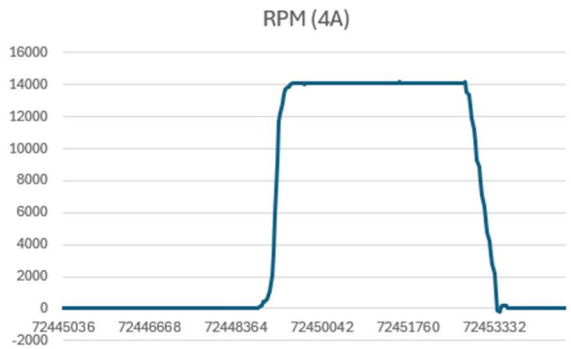

Ampere (A)
Duty Cycle
Seconds (s)
Current in
Current motor
Duty cycle

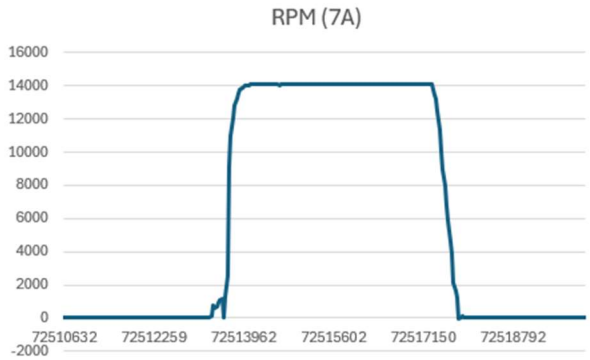
15A

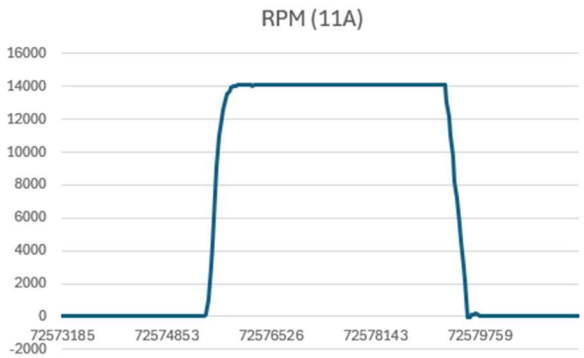

Ampere (A)
Duty Cycle
Seconds (s)
Current in
Current motor
Duty cycle

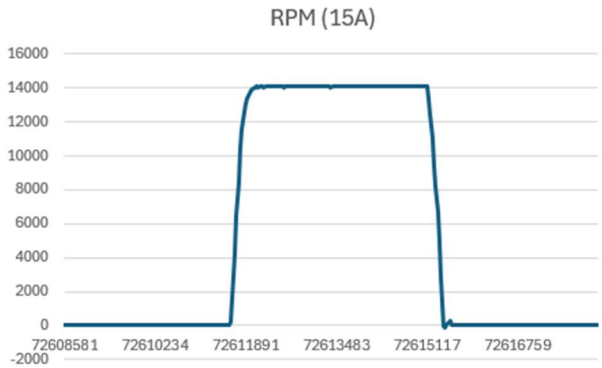

RPM (STOP)
16000
14000
12000
10000
8000
6000
4000
2000
0
72301082
72302667
72304293
72305886
72307447
72309080

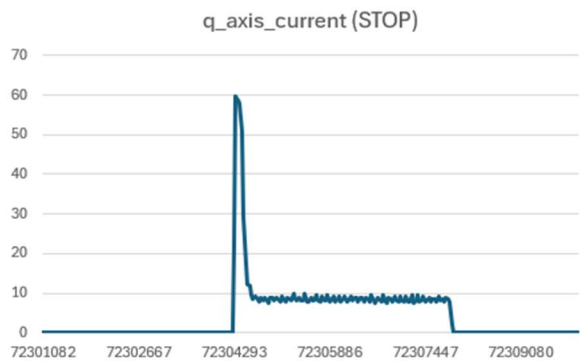

RPM (1A)
16000
14000
12000
10000
8000
6000
4000
2000
0
-2000
72376715
72378313
72379993
72381619
72383257
72384876
72386527

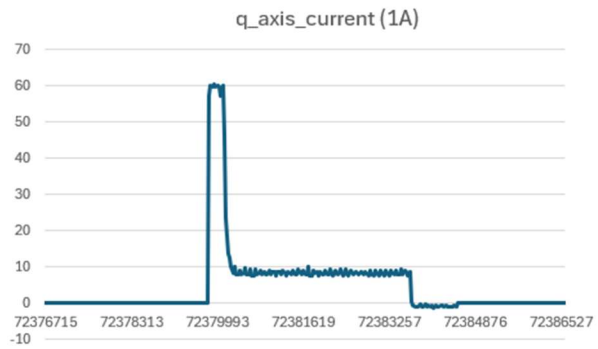

RPM (4A)
16000
14000
12000
10000
8000
6000
4000
2000
0
-2000
72445036
72446668
72448364
72450042
72451760
72453332

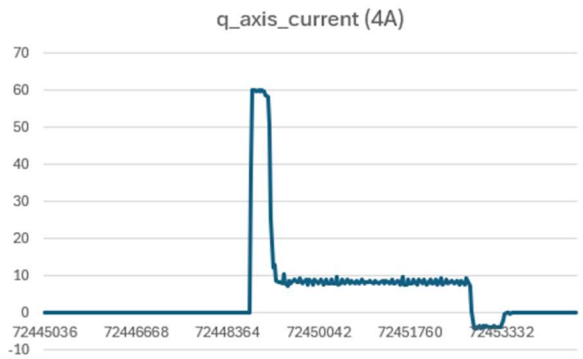

RPM (7A)
16000
14000
12000
10000
8000
6000
4000
2000
0
-2000
72510632
72512259
72513962
72515602
72517150
72518792

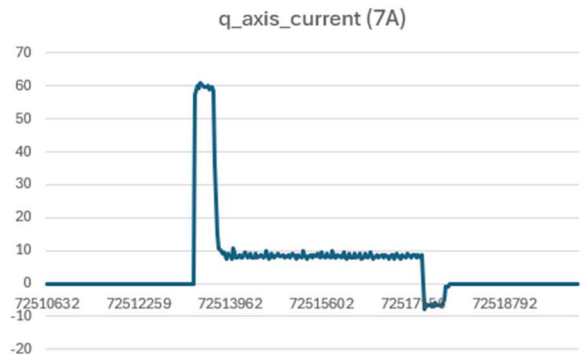

RPM (11A)
16000
14000
12000
10000
8000
6000
4000
2000
0
-2000
72573185
72574853
72576526
72578143
72579759

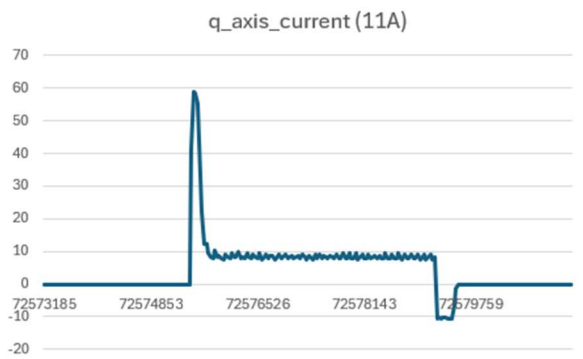

RPM (15A)
16000
14000
12000
10000
8000
6000
4000
2000
0
-2000
72608581
72610234
72611891
72613483
72615117
72616759


q_axis_current (STOP)
70
60
50
40
30
20
10
0
72301082
72302667
72304293
72305886
72307447
72309080


q_axis_current (1A)
70
60
50
40
30
20
10
0
-10
72376715
72378313
72379993
72381619
72383257
72384876
72386527


q_axis_current (4A)
70
60
50
40
30
20
10
0
-10
72445036
72446668
72448364
72450042
72451760
72453332


q_axis_current (7A)
70
60
50
40
30
20
10
0
-10
-20
72510632
72512259
72513962
72515602
72518792


q_axis_current (11A)
70
60
50
40
30
20
10
0
-10
-20
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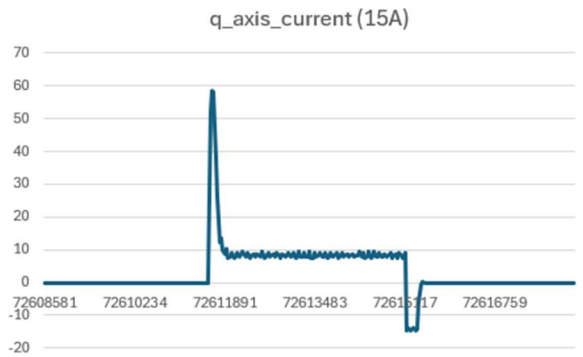
72574853
72576526
72578143
72579759


q_axis_current (15A)
70
60
50
40
30
20
10
0
-10
-20
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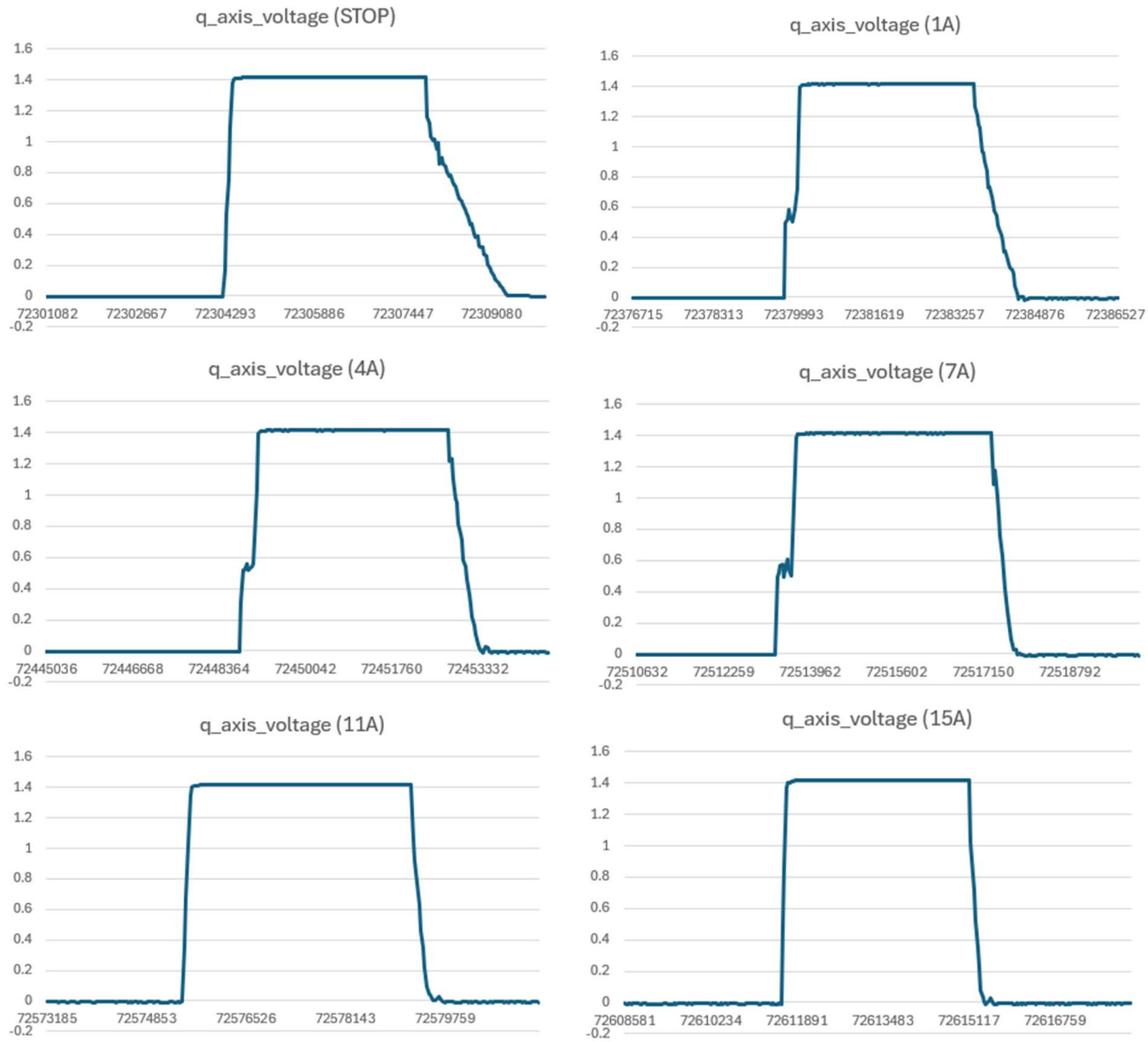
72611891
72613483
72616759

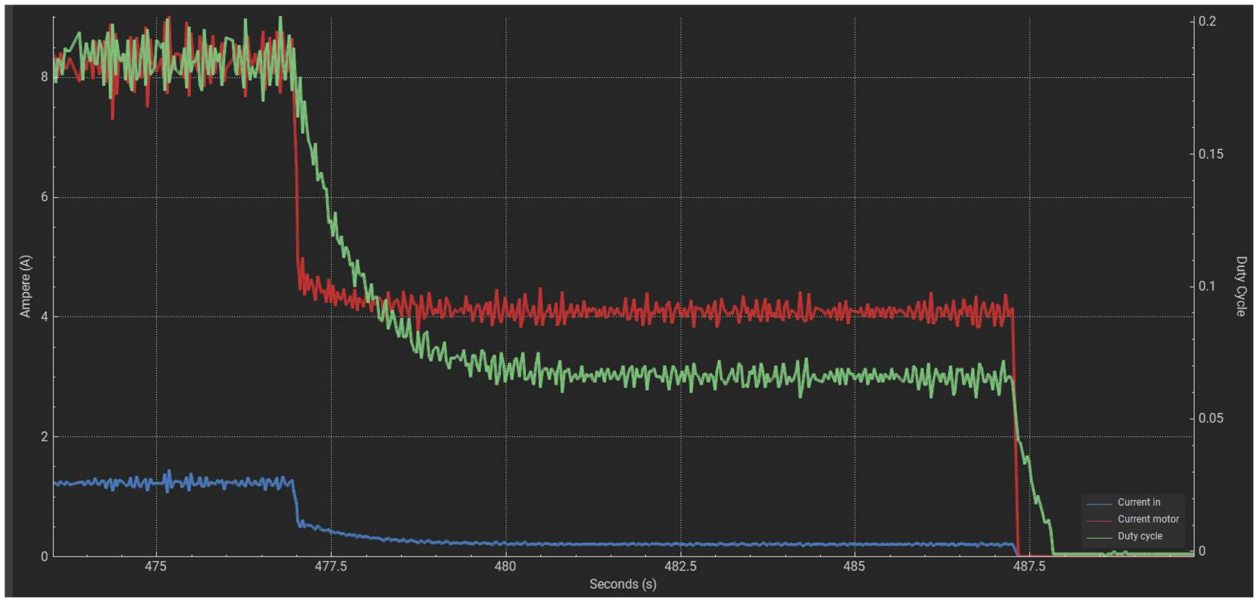


When the motor is stopped using the standard **STOP** command, the speed and q-axis voltage waveforms exhibit a gradual decay as the motor coasts to a halt. In contrast, applying a brake current I_B result in a significantly steeper deceleration slope, where increasing the current from 1A to 15A proportionally increases the rate of deceleration. This behavior is driven by the relationship between torque and q-axis current by commanding a negative I_B, the VESC generates a counter-torque that actively opposes the motor's rotation.

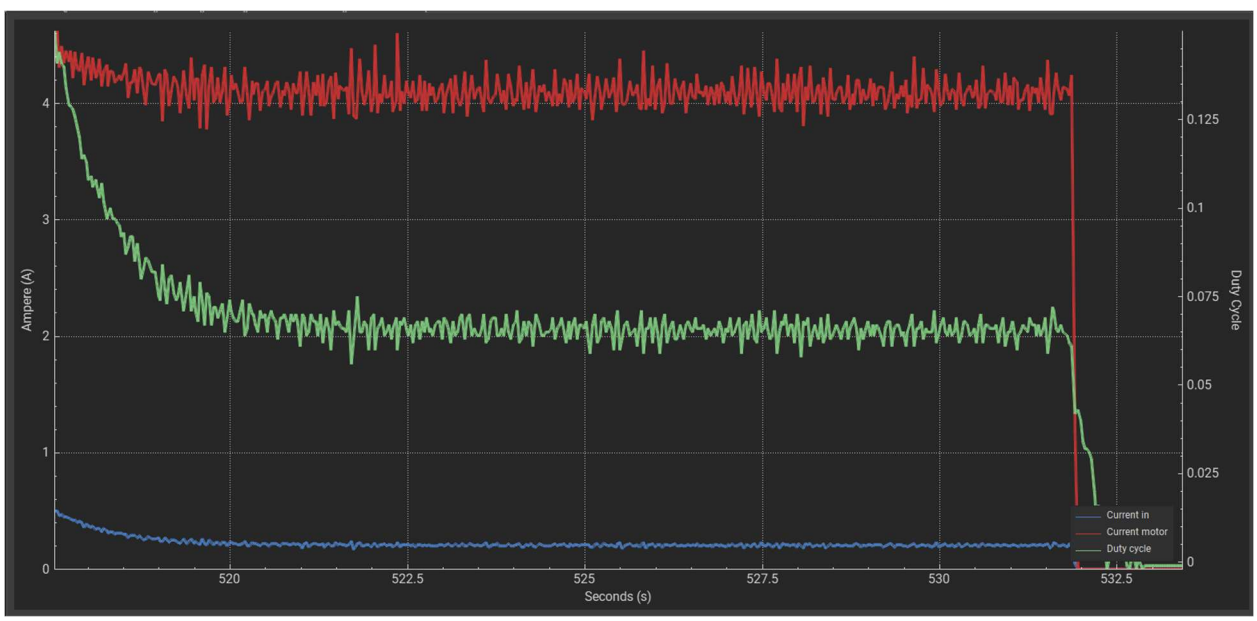
The current waveforms further illustrate this principle: while a normal stop simply causes the current to drop to zero, an active brake command produces a negative spike in the q-axis current. As I_B is increased, this negative current magnitude grows, effectively converting the motor's kinetic energy into electrical energy more rapidly through the principle of electromagnetic braking. Consequently, the higher the negative q-axis current, the more aggressive the counter-torque, leading to the rapid decline in both RPM and voltage observed in the steady-state to stop transition.

Question 4: Conduct an experiment where you apply a current command. Apply a current command of 8A. Note that you should not apply more than 8A to the motor at no load condition. Record the real-time speed, current, duty cycle, and voltage data. When the motor speed is at steady state, reduce the current command to 4A. Record the same data. Note down the speed at steady-state. Explain how the speed, current, voltage, and duty cycle changed. Stop the motor. Apply a constant current command of 4A. Record the waveforms and explain the operation of the motor. What happened? Stop the motor. Apply a speed command for the speed you noted down earlier. Record the data and compare the results with those from the previous experiments. How did the motor operation change? Why?

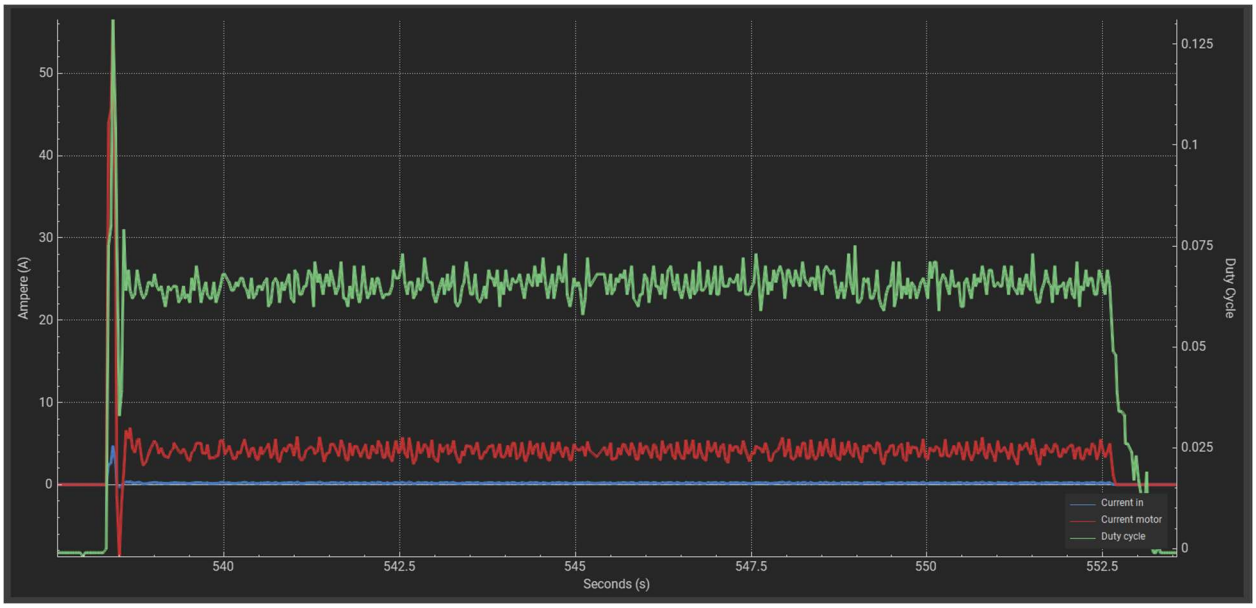
8A -> 4A



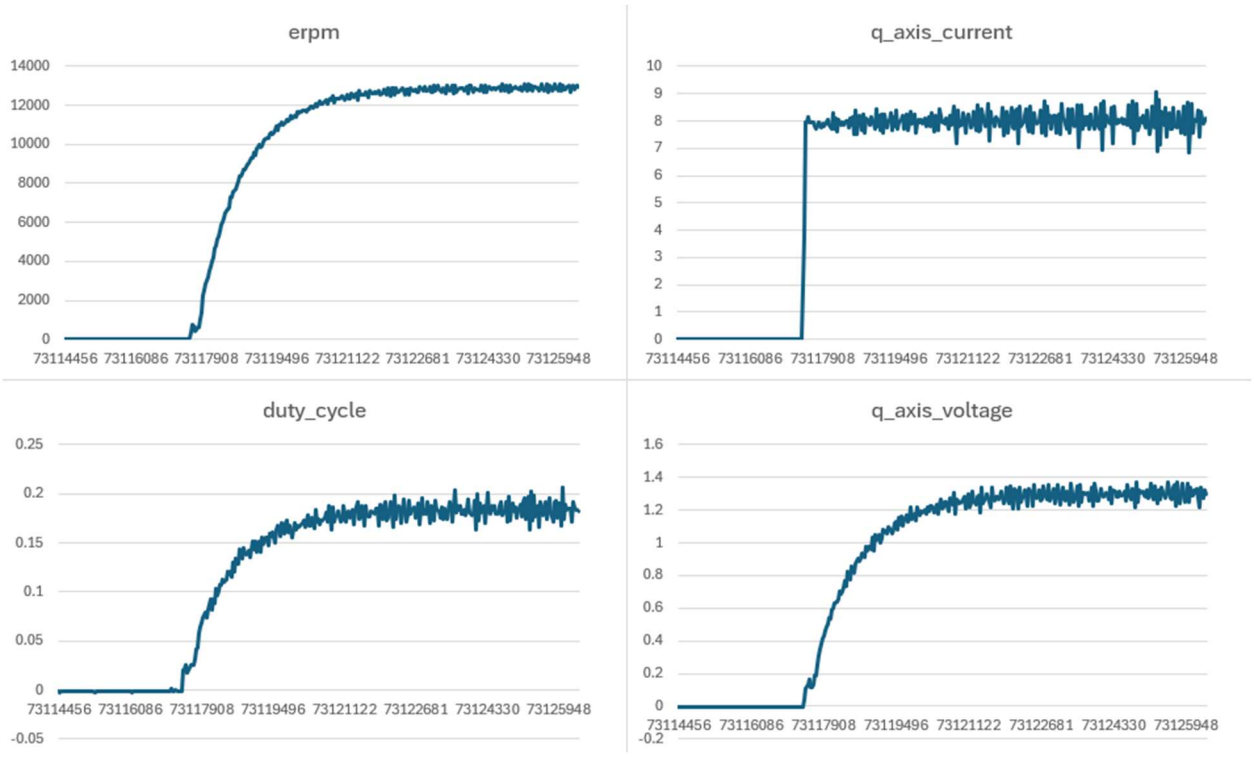
4A



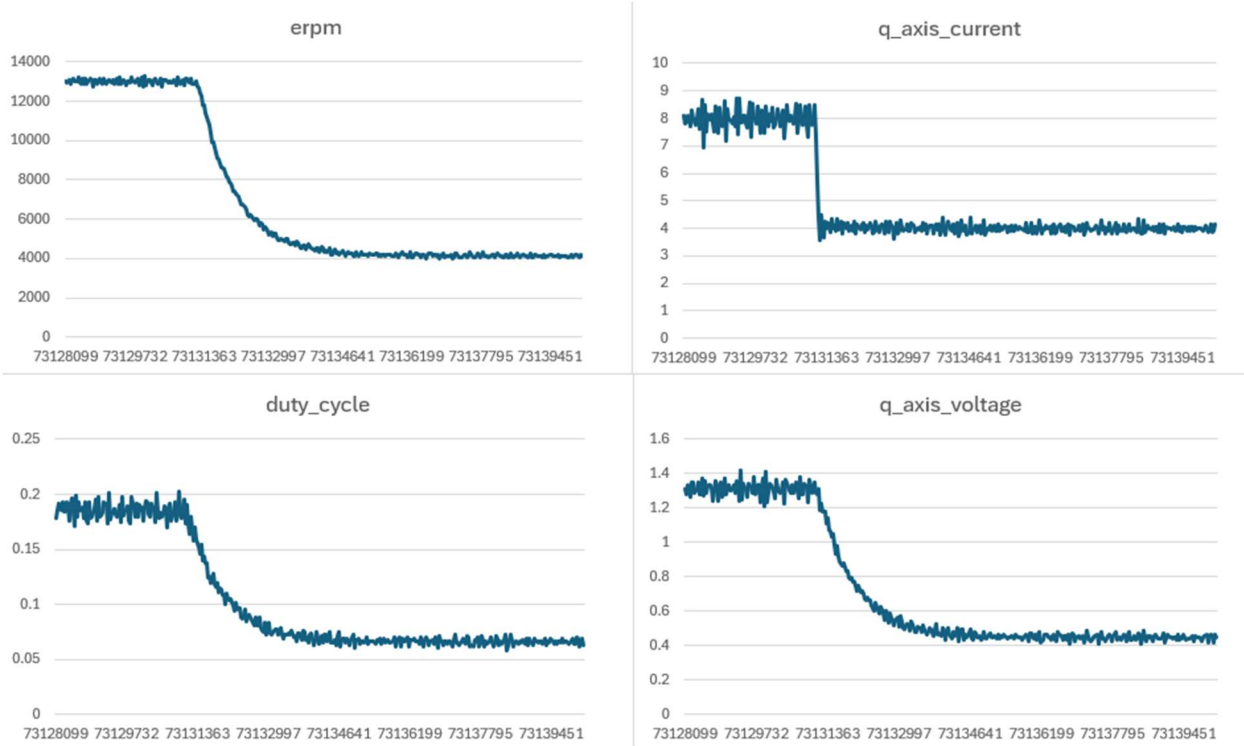
Speed at 4A is around 4100 RPM



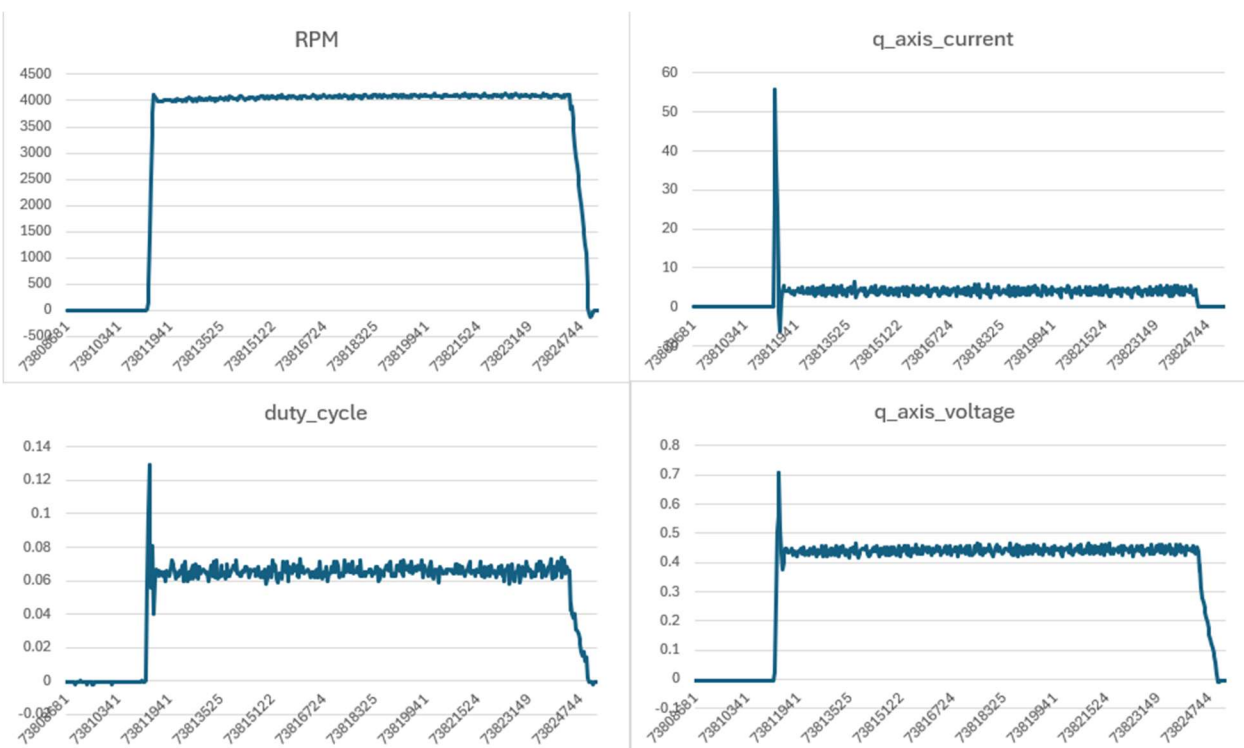
At 8A



From 8A -> 4A



At 4100 RPM

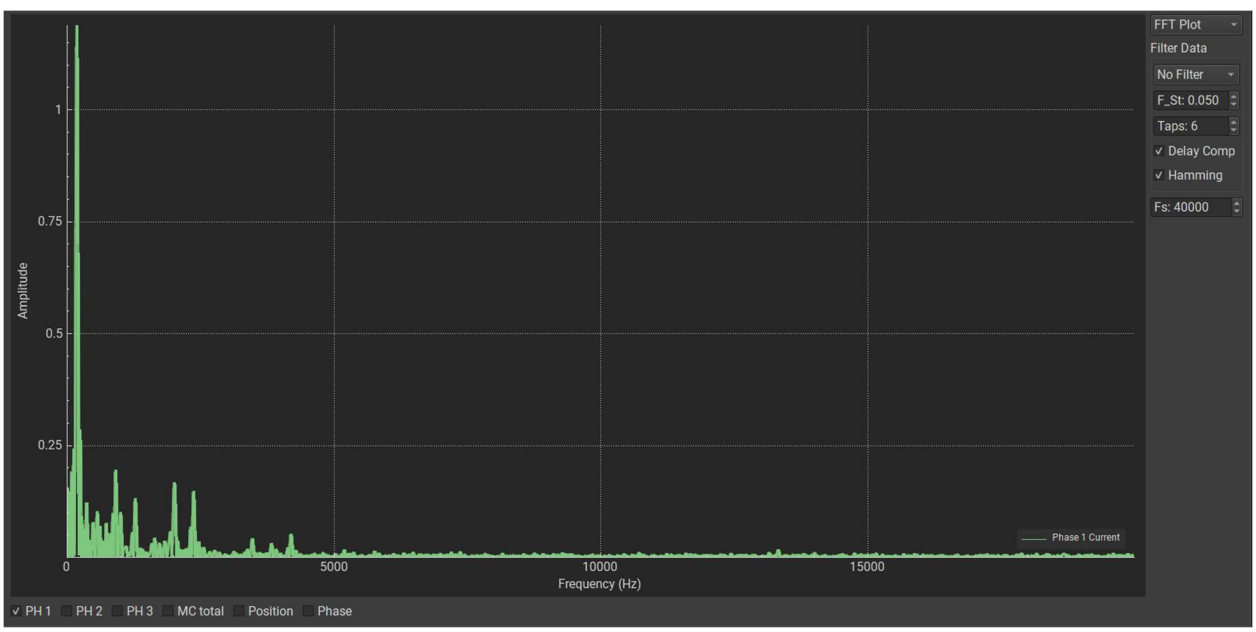


In this experiment, the transition from an 8A current command to a 4A current command demonstrates the motor's response to a reduction in electromagnetic torque. At the initial 8A command, the motor accelerates until the torque matches the friction and windage losses at approximately 13,000 RPM. When the command is reduced to 4A, the motor begins a steady deceleration because the newly reduced torque is no longer sufficient to maintain the higher speed. This deceleration continues until the motor settles at a lower steady-state speed of 4100 RPM, where the torque produced by 4A once again balances the friction losses. During this phase, both the duty cycle and voltage decrease proportionally with the speed, as the VESC requires less voltage to push the same current through the motor as the Back EMF drops. Same behaviour with 0 to 4A, but it is accelerating.

This contrasts sharply with the Speed Command (4100 RPM), where the VESC utilizes its PID controller to close the error gap instantly. While the 4A current command results in a gradual, torque-limited acceleration or deceleration toward the equilibrium point, the speed command triggers an immediate current spike to reach the target velocity as fast as possible. This can be seen by the steep slope at the RPM. Once at steady state, the speed command only draws the minimal current necessary to overcome friction, whereas the 4A current command maintains a constant torque output regardless of the motor's actual needs.

<u>Objective 3:</u>

<u>Question 5:</u> Explore the Sampled Data feature in the VESC Tool. For constant speed operation in FOC mode, record the sampled phase current waveforms. Explore the FFT option in the VESC tool and apply it to the phase current measurements. Report the results by applying the speed command and then click on Sample Now button. Attach a screenshot of your FFT plot in your report.

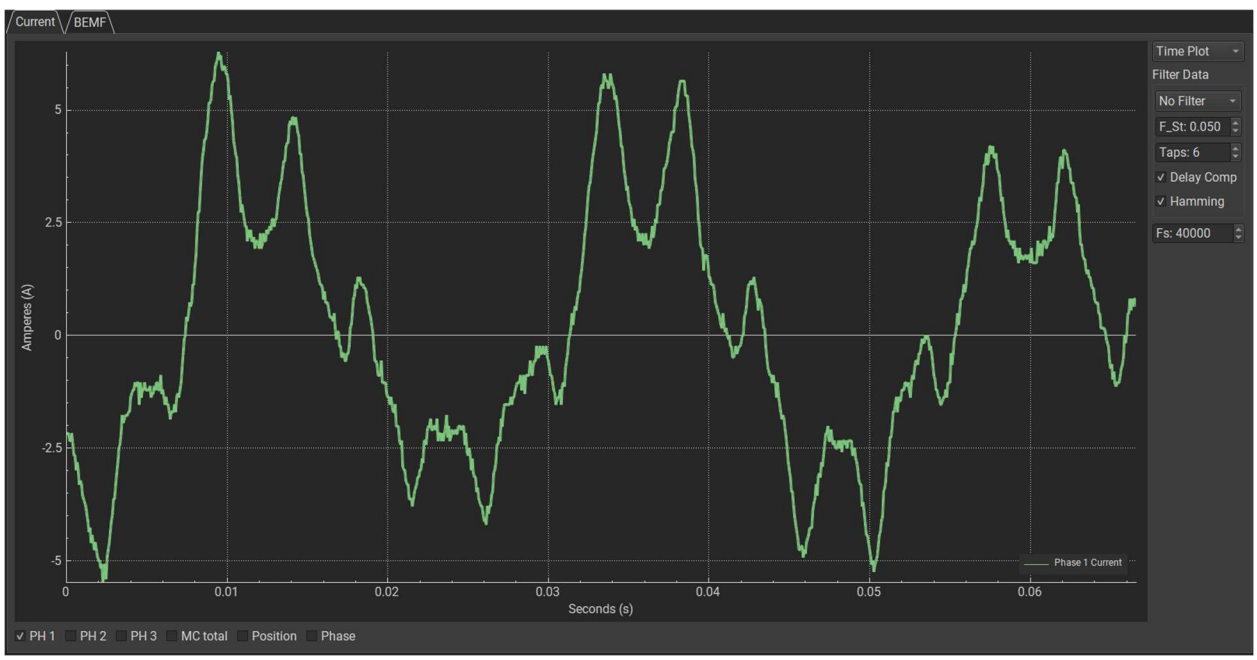


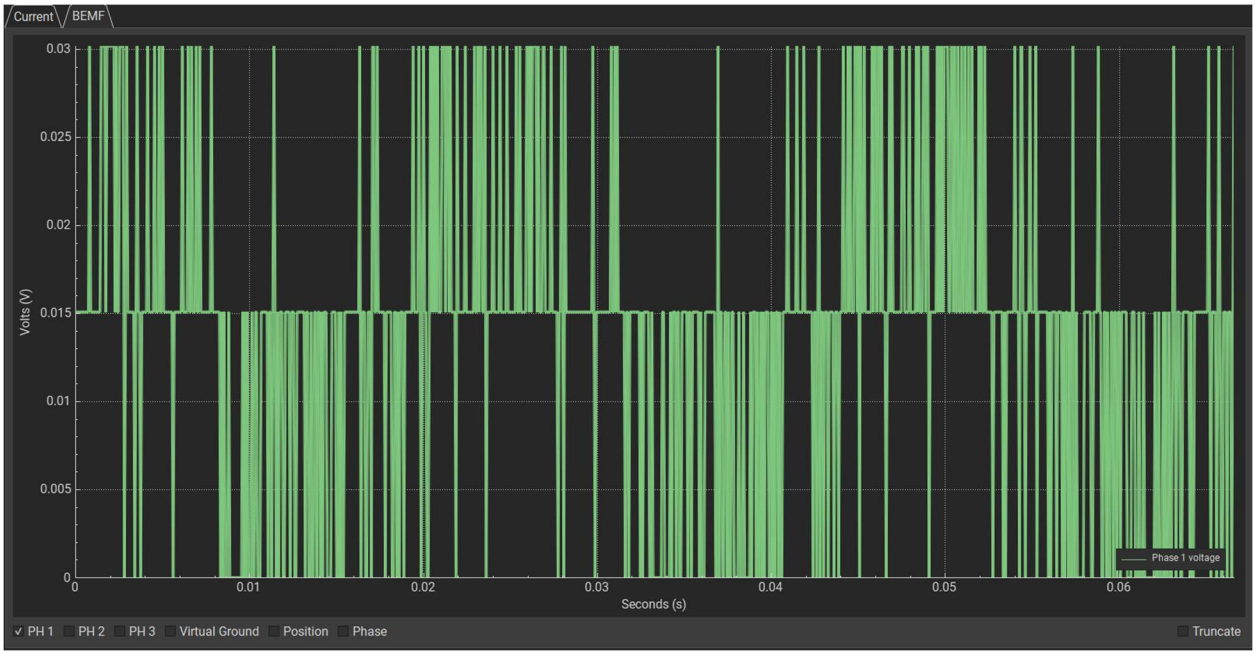
Question 6: What is an FFT plot? Why is an FFT plot applied to the phase current measurements important when analyzing the behavior of a motor

A FFT (Fast Fourier Transform) plot serves as a graphical representation of a signal within the frequency domain. It is produced by performing a Fourier Transform on a time-domain signal, effectively decomposing the original waveform into its individual sinusoidal components. In this visualization, the x-axis denotes the frequency of the signal measured in Hertz (Hz), while the y-axis indicates the corresponding amplitude or, depending on the application, the power of those frequency components.

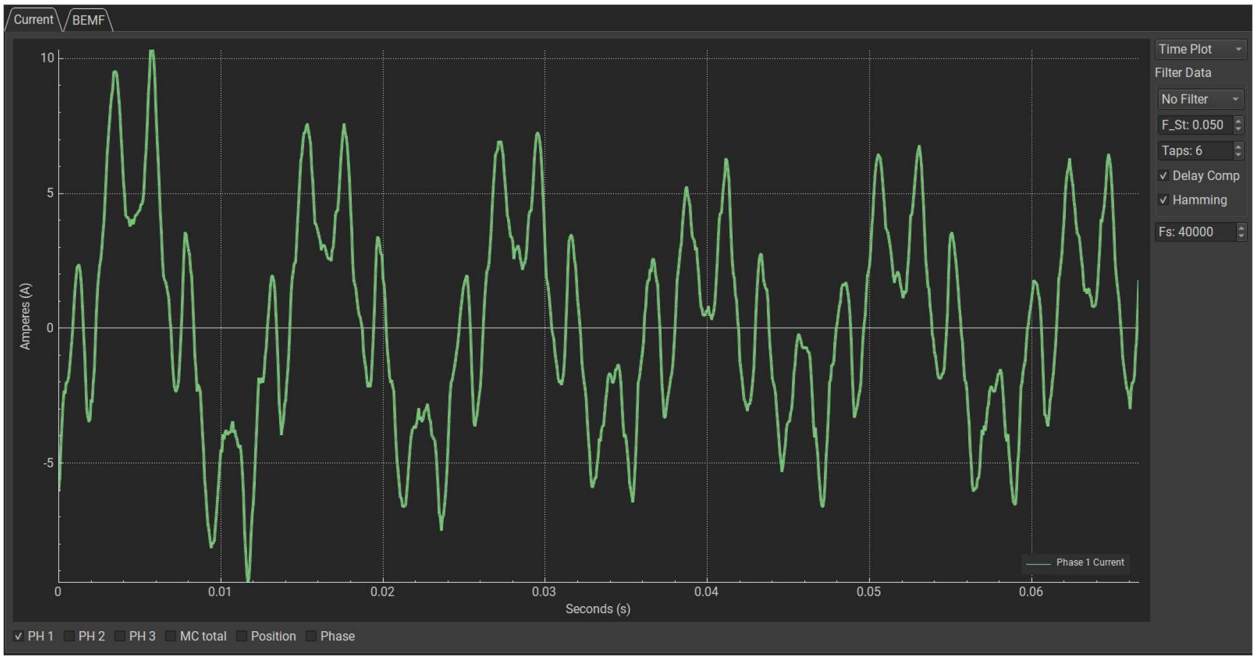
Question 7: For 2500 rpm, 5000 rpm, 7500 rpm, record the motor phase currents and phase back- emfs using the Sampled Data feature. For each speed and each phase, plot the phase current and phase back-emf on the same graph. You might need to adjust the scale of the current and voltage to show them in the same graph. What is the phase shift between the phase current and phase back-emf at each speed and for each phase? How does the frequency of the phase current and phase back-emf change with speed? Explain the reasons using the electric motor fundamentals

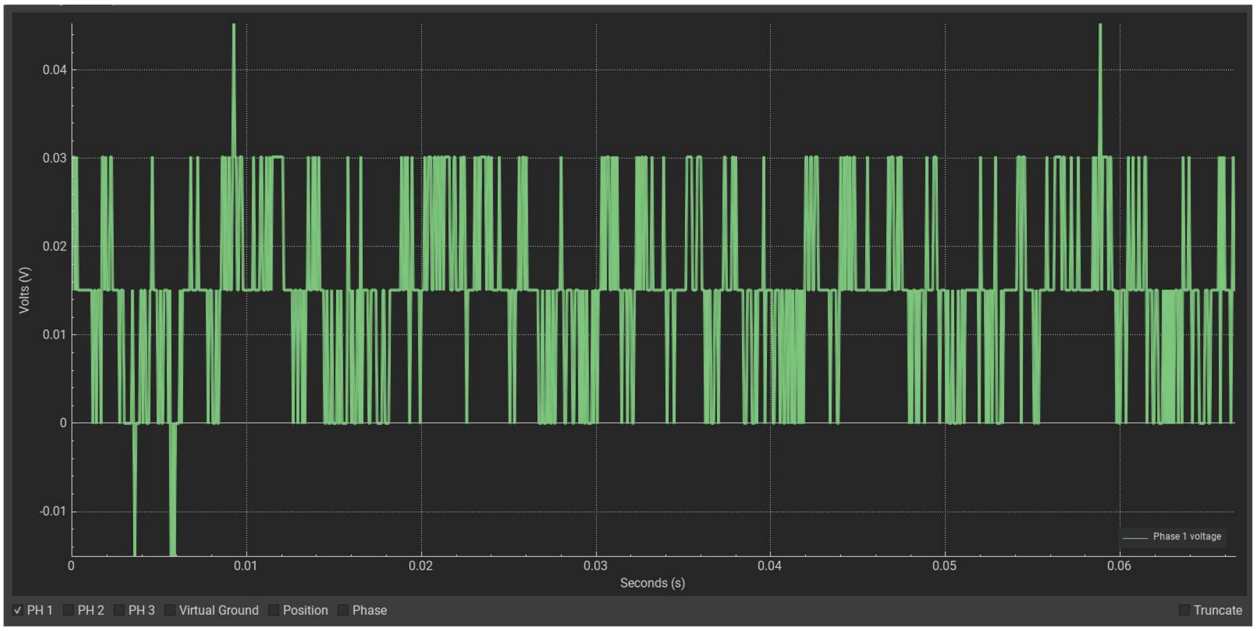
2500



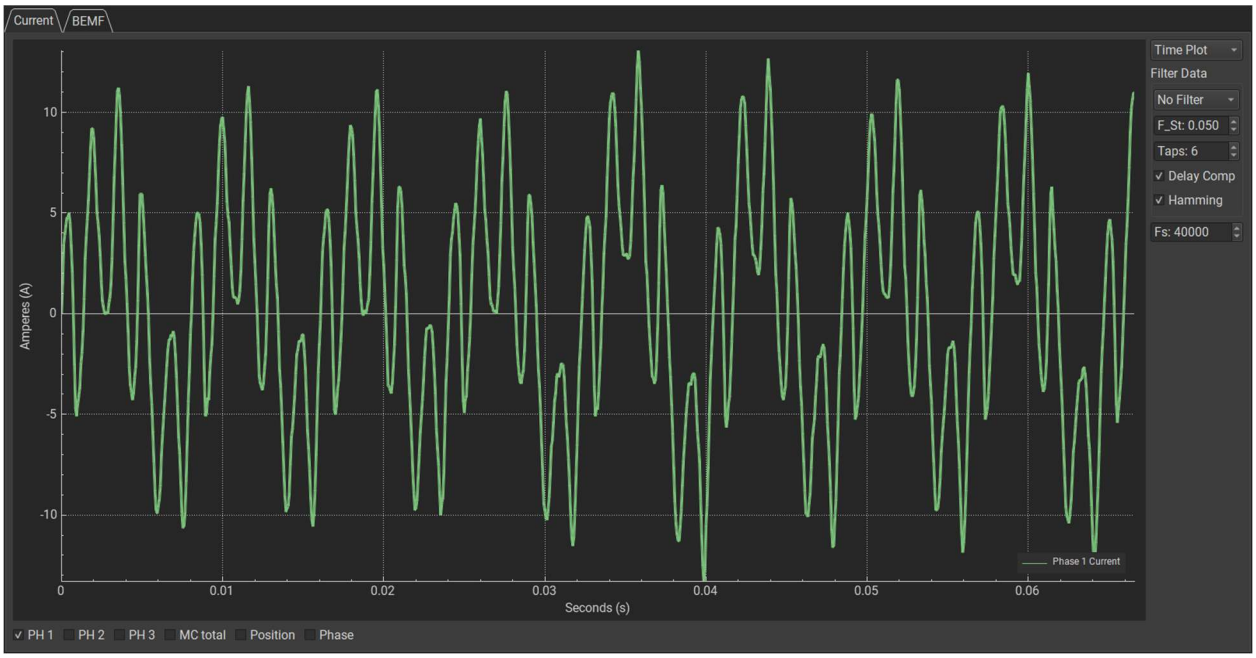

Current
BEMF
Volts (V)
0.03
0.025
0.02
0.015
0.01
0.005
0
0
0.01
0.02
0.03
0.04
0.05
0.06
Seconds (s)
Phase 1 voltage
PH 1
PH 2
PH 3
Virtual Ground
Position
Phase
Truncate

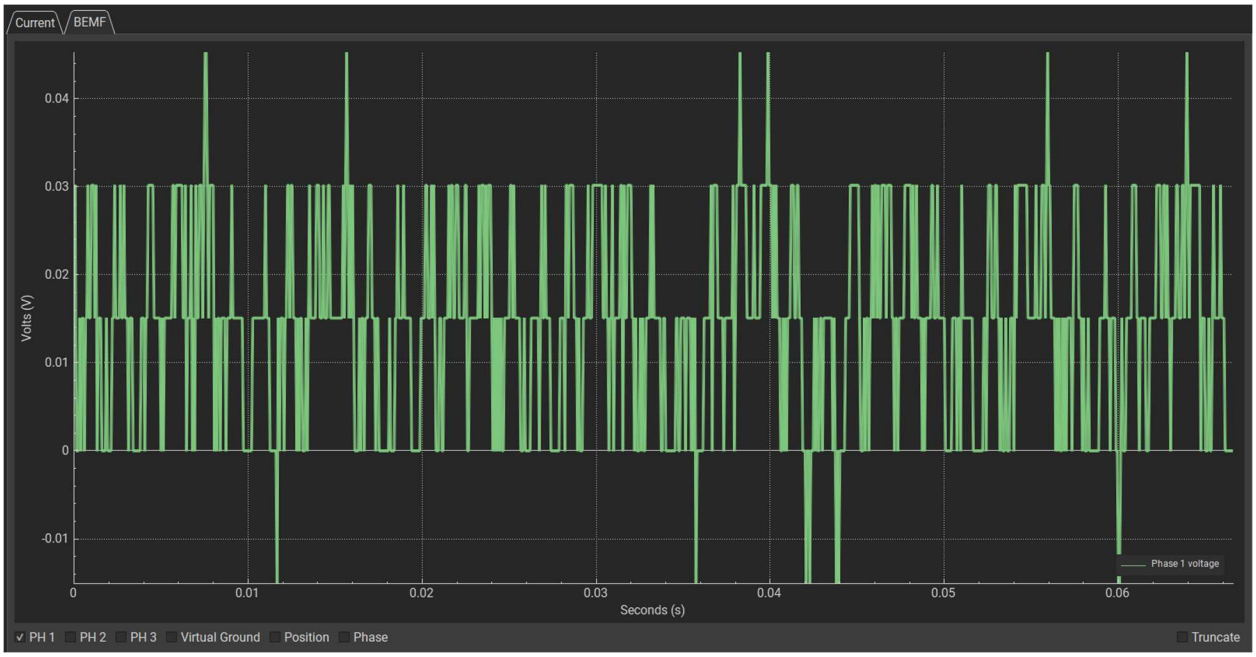
5000

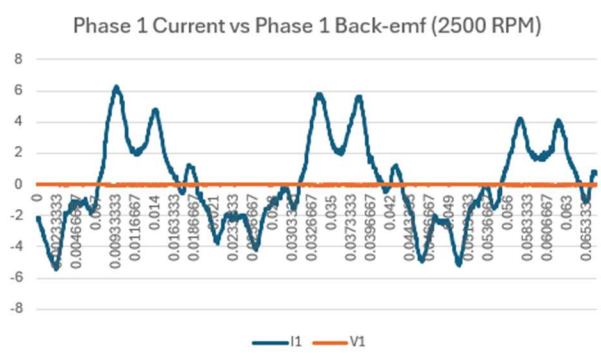

Current
BEMF
Amperes (A)
10
5
0
-5
0
0.01
0.02
0.03
0.04
0.05
0.06
Seconds (s)
Phase 1 Current
Time Plot
Filter Data
No Filter
F_St: 0.050
Taps: 6
Delay Comp
Hamming
Fs: 40000
PH 1
PH 2
PH 3
MC total
Position
Phase

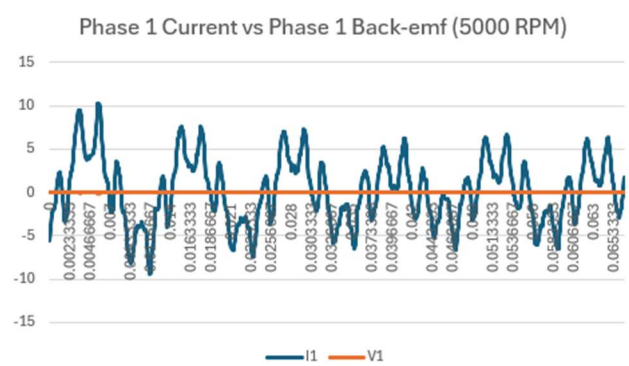

Volts (V)
0.04
0.03
0.02
0.01
0
-0.01
0
0.01
0.02
0.03
0.04
0.05
0.06
Seconds (s)
Phase 1 voltage
PH 1
PH 2
PH 3
Virtual Ground
Position
Phase
Truncate

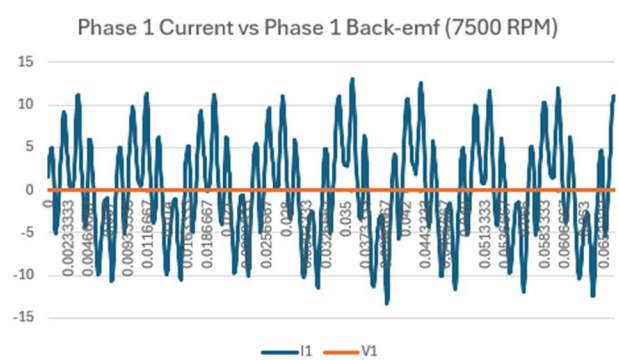
7500


Current
BEMF
Amperes (A)
10
5
0
-5
-10
0
0.01
0.02
0.03
0.04
0.05
0.06
Seconds (s)
Phase 1 Current
Time Plot
Filter Data
No Filter
F_St: 0.050
Taps: 6
Delay Comp
Hamming
Fs: 40000
PH 1
PH 2
PH 3
MC total
Position
Phase









For all three speeds, the measured phase currents showed the expected waveforms. Ideally, we should see that the phase current and phase back EMF for each phase should be approximately in phase in a controlled environment. However, the graph for the back EMF waveform did not match the expected sinusoidal shape, and instead appears inconsistent, indicating a measurement or sampling error. Some possible reasons for this could be due to improper scaling or sampling

timing issues. This error makes the experimental phase shift difficult to accurately record and analyze; therefore, theoretical expectations will be used for analysis.

As the motor speed increases, the frequency of both the phase current and back EMF also increases proportionally. This is because the frequency is directly related to the RPM and the number of pole pairs. As the RPM increases, the magnetic flux changes faster, producing higher voltages and current. The back EMF also increases with speed due to the relationship between back EMF and angular velocity, while the is current adjusted due to the torque. Overall, despite the incorrect back EMF measurement, using motor fundamentals we can reason that the phase current and back EMF should be aligned in phase, while the frequency increases linearly with speed due to faster motor rotation and changing magnetic fields.